ARTICLE

PRISM: a modular system for glioma progression surveillance on MRI

Sri Datta Ganesh Bandreddi[a] and Jessica Zhang[b]

[a]Department of Biomedical Engineering, Carnegie Mellon University, Pittsburgh, PA, USA; [b]Department of Mechanical Engineering, Carnegie Mellon University, Pittsburgh, PA, USA

ARTICLE HISTORY

Compiled October 26, 2026

ABSTRACT

Surveillance MRI after treatment for glioblastoma and other aggressive gliomas must be interpreted with prior imaging and the treatment course. A progression score alone does not show the clinician the evidence that produced it or what it leaves out. We built PRISM, a modular system that combines image-derived measurements, treatment context, separate image and timeline pathways, uncertainty and provenance in a locally generated, checked report. On MU-Glioma-Post, the timeline and image pathways reached AUCs of 0.82 and 0.71. Applied unchanged, the combined score reached 0.77 on LUMIERE; at UCSF-ALPTDG, single-scan image measurements reached 0.80 and the image pathway 0.73. The evidence behind the scores differed by cohort: treatment timing at MU-Glioma-Post and enhancing volume at UCSF-ALPTDG. In a second, cross-cohort report evaluation, 406 of 406 reports passed a lexical check against their structured records. PRISM is designed to make evidence inspectable, not to replace clinical judgement; its effect on clinical decisions needs a reader study.

KEYWORDS

glioma surveillance; external validation; clinical decision support

1. Introduction

Glioblastoma and other aggressive gliomas require repeated MRI surveillance after surgery and chemoradiotherapy. At each scan, the clinical team interprets the current images against earlier images and the course of treatment. The decision is hardest soon after radiotherapy, when treatment can increase contrast enhancement without tumour growth. This pseudoprogression is most common in the first weeks after chemoradiotherapy and more frequent when the promoter of MGMT, a DNA-repair gene, is methylated (Taal et al. 2008; Brandes et al. 2008). The information needed for that judgement is longitudinal: images and their change, treatment timing and the parts of the record that are known on the scan day.

Radiomics and other prediction models can quantify image patterns, but a probability alone is not a surveillance summary. It does not show the treatment context, the measurements behind an image input, the uncertainty around a score, or what the model did not use. These omissions

CONTACT Sri Datta Ganesh Bandreddi. Email: sbandred@andrew.cmu.edu; ORCID: 0009-0004-3617-6734

matter because public surveillance datasets often contain images but few treatment dates; centres document progression differently; the hardest situations are uncommon; and segmented volumes need to be inspected rather than accepted as a hidden input. Small cohorts also make a single test split an unstable account of performance (An et al. 2021; Varoquaux 2018). Only 3 of 20 studies in a recent systematic review of automated response assessment included external evaluation (Shi et al. 2025).

Our initial radiomics work on MU-Glioma-Post motivated a broader question: can a system turn the evidence available at a post-treatment scan into a report that a clinician can inspect? We asked whether PRISM could retain image and treatment-timeline evidence separately, evaluate those components at other centres without refitting, and state where the evidence is thin, while writing the resulting report inside the hospital.

PRISM is that system. It turns each scan into a structured evidence record: clinical and treatment-timeline inputs, measurements from the segmented tumour, separate image and timeline scores, a combined score, uncertainty fields and their provenance. A local language model writes the narrative from that record, and a deterministic checker tests the narrative against it. We developed PRISM on MU-Glioma-Post (MU) and tested locked models on LUMIERE and UCSF-ALPTDG (UCSF). The component models ranked scans at the external centres, but the evidence carrying a score differed by cohort. The report makes that distinction visible; its lexical grounding and required content were tested, but no clinician reader outcome was measured.

2. Methods

2.1. PRISM

PRISM handles each scan in five modules (Figure 1) and writes their outputs to one structured evidence record. The timeline pathway and the image pathway keep the two kinds of evidence apart, so that the image pathway can run where no treatment dates exist and the audit can show which evidence a score used. Segmentation masks supply the measurements and images displayed in the report. The record also carries a set of plausible outcomes, agreement between pathways, how many scans like this one have been tested and the provenance of every displayed value. The report is written locally from that record, so patient data do not leave the hospital.

The timeline pathway scores a scan from clinical data and the patient's position in treatment. On MU its inputs are age, sex, IDH mutation and MGMT promoter methylation, which alone form the clinical-only reference model, together with the days since diagnosis and since surgery, the start, end and dose of radiotherapy, chemotherapy and the visit index, each taken as of the scan day. Where a cohort records no treatment dates, PRISM uses a reduced timeline that keeps the clinical inputs, the days since surgery and a flag for the first weeks after radiotherapy. The flag comes from the rater's note at LUMIERE and is approximated from the start of radiotherapy at UCSF. Both pathways are ridge logistic regressions; scaling and any selection were fitted on the training data only. An unknown IDH or MGMT status is coded as the reference category, and a missing measurement is filled with the training median, with a missing-value indicator, inside the fitted pipeline.

The image pathway scores a scan from simple measurements of the segmented tumour. On MU its 27 inputs are the volumes, shape and relative intensity of the segmented compartments at the

scan, their change from the previous scan, and the enhancing and total lesion volumes relative to the smallest in the patient's history ([Supplementary section: image pathway inputs and their harmonisation at LUMIERE and UCSF]). It holds no dates, intervals or treatment flags, because such inputs carry when a scan was taken, which the timeline already holds, and the image pathway has to run where no dates exist (Section 3.3). A single-scan version keeps only the 19 measurements of the current scan.

The combined score is one ridge model fitted on the inputs of both pathways and is shown beside them. At LUMIERE and UCSF its inputs are the reduced timeline, the harmonised volumes and their change, the interval since the previous scan and whether there was one.

The measurements come from the segmentation masks supplied with each cohort, which are the automatic masks released with MU (Mahmoud et al. 2025), the DeepBraTumIA masks of LUMIERE (Suter et al. 2022) and the masks provided with UCSF (Fields et al. 2024). PRISM renders the report's figure from the same masks by code.

Each scan's confidence has three parts, a conformal set, a pathway-agreement statement and a reliability card (Mitchell et al. 2019). The conformal set is a patient-level split conformal prediction set at the 80% level (Angelopoulos and Bates 2023), built to contain the true outcome for at least 80% of scans; it holds one outcome when the score is clear and both when it is not. It comes from the combined model refitted on 70% of the MU patients and calibrated on the remaining 30% ([Supplementary section: system results]); on MU this split is made inside each training fold, and for the external cohorts one MU-calibrated set was applied unchanged, which assumes that their scans are exchangeable with MU's. The two pathways agree when both scores lie on the same side of that pathway's median score over the MU training scans of the locked models (on MU, over the held-out scores of the other folds), so the rule can be applied at a single scan; it uses scores only, never labels. The reliability card looks up the scan's situation, its phase of care crossed with the change in enhancing tumour since the previous scan, in a table of each cohort's own held-out results, which include the scan being looked up. A situation is tested, and the card gives its AUCs and intervals, when it holds at least 10 progression and 10 non-progression scans; it has few cases, and the card gives counts only, when it holds fewer; and it is not tested when it holds none.

A local language model writes the report from the scan's record, and a deterministic checker tests the report against that record. The model (Qwen3.8-27B) runs offline on one GPU at temperature 0 and receives only the record; the checker rejects any number, flag or confidence statement that is not in the record, any omission of required content, and any wording that offers a diagnosis, advice or a probability (Supplementary Section S19).

2.2. Cohorts and the task

PRISM was developed on MU and tested on LUMIERE and UCSF, in each case on grade 4 glioma as the primary population, with models fitted once on MU and applied unchanged, which we call locked. Table 1 lists what each cohort records; UCSF holds one pair of consecutive post-treatment scans per patient, and because the first scan has no earlier scan we evaluated the second only. Progression is documented in each cohort's own record, as a dated clinical decision at MU, as an expert rating of response at each scan at LUMIERE and as a progression date supplied with the dataset at UCSF.

The primary task is to decide, at each scan, whether this is the scan at which progression is

documented. On MU a positive is the first scan on or after the documented progression date, and a negative is a scan followed by at least 120 days without progression. At LUMIERE a positive is the first scan rated as progressive disease, and a negative is an earlier scan rated stable or responding. At UCSF a positive is a second scan with progression dated within 30 days of it and nearer to it than to the first scan; a negative is a second scan with progression at least 120 days later, or with at least 120 days of follow-up without progression. On MU the intended use is scans from the start of radiotherapy onwards, and unless stated MU results are for this population; LUMIERE and UCSF are evaluated on all scans in their detection sets. We divide the intended use into three phases of care, which are during radiotherapy, early after radiotherapy (up to 84 days, 12 weeks, after its end) and later.

2.3. Evaluation

On MU, every model was evaluated by repeated patient-grouped cross-validation. We used five repeats of five folds, with all scans of a patient in the same fold, and report the mean over repeats of the pooled held-out AUC with a 95% confidence interval (CI) from 1,000 patient-level bootstrap draws. For the models and the conformal sets, no patient appears in more than one of fitting, calibration and testing. We chose repeated splits because in our first evaluation a radiomics model reached an AUC of 0.80 on one test split and averaged 0.72 when re-selected and re-tested over 500 splits of the same design (Supplementary Section S8 and Figure S2).

For the external tests, the locked models, fitted on all grade 4 detection scans of MU from every phase of care, were applied under plans committed before any model score was compared with an external label; the one deviation from a plan is in the note to Table 2 ([Supplementary section: deviations from the plans]). External AUCs have 95% CIs from 1,000 patient-level bootstrap draws. As a reference, we also fitted each UCSF model on UCSF itself by repeated cross-validation. Table 2 lists the questions fixed in advance and their answers.

Results of analyses not fixed in advance carry a dagger (†) where they first appear in a paragraph, legend or table, and the analyses are listed in [Supplementary section: analyses not fixed in advance]. Among them are the definition of the image pathway on MU, whose use at UCSF and in the system tests was planned, and the agreement rule and the reliability card, which were defined after the cohort results had been seen. Comparisons within a phase of care or within an enhancement category serve as diagnostics of what a score depends on.

3. Results

The initial radiomics exercise showed why a progression model could be useful, but not what a clinician should see at a surveillance scan. PRISM therefore treats a score as one field in a longitudinal evidence record, alongside the measurements, treatment context, uncertainty and the inputs not available to the model. We first evaluate the two model pathways, then test whether their behaviour transfers and whether the report faithfully presents the resulting record.

3.1. The pathways provide distinct evidence at the development centre

In the intended use on MU (146 scans; Table 1), the image pathway† detected the progression scan with an AUC of 0.71 (95% CI 0.63–0.78), and the timeline pathway with 0.82 (0.75–0.90), mostly by separating phases of care, within which it did no better than the images (Section 3.3). Clinical data alone, without the timeline, scored 0.53, close to chance. The combined score that PRISM shows, fitted on the inputs of both pathways†, also reached 0.82, so on MU the images did not raise the timeline's AUC. The imaging model fixed in advance, a different model that adds measurements and three schedule inputs to the timeline (Section 3.3), fell short of its pre-specified gain of 0.03 (Table 2). Predicting progression within 120 days of a scan was harder; the model reached 0.60 (0.52–0.69) in the intended use (Supplementary Section S23). A timeline pathway needs treatment dates, which the two external cohorts largely lack, so we next tested how each pathway transfers to them.

3.2. The components rank scans at two external centres without refitting

The locked combined model reached 0.77 (0.68–0.86) on LUMIERE (Figure 2) and met its planned transfer rule (Table 2). At LUMIERE the combined score was above either pathway alone, images 0.65 and the reduced timeline 0.69† (the gain over the timeline alone had an interval that included zero). Clinical data alone stayed close to chance in all three cohorts (Figure 2). Enhancing tumour volume alone, with no fitted model†, scored 0.62 on MU, below the image pathway, and 0.66 on LUMIERE, level with it; at LUMIERE only the combined score did better than this simplest baseline (Figure 2).

At UCSF, both planned image models transferred. Measurements of the single scan reached 0.80 (0.69–0.89), and the image pathway, which adds change from the previous scan, reached 0.73 (0.60–0.84), meeting its planned transfer rule (Table 2). The change inputs therefore transferred less well than the measurements of the scan itself. Training at UCSF raised the image pathway to 0.86 (difference 0.14, 0.03–0.26), while the single-scan model scored about the same when trained there†. The reduced timeline, whose flag for the early weeks after radiotherapy UCSF can only approximate (Section 2.1), had the lower AUC of the two pathways, 0.56. The locked combined score reached 0.86 (0.76–0.93), but at UCSF its timeline inputs are approximated, and part of its AUC comes from the interval between the two scans, which was shorter before progression† (Figure 2).

These AUCs describe how the scores rank scans. Read as a risk, the combined score was calibrated in the large at LUMIERE† but overpredicted progression at UCSF (observed over expected 0.72, 0.55–0.91), and a threshold set at the MU median† called 90% of UCSF scans positive, so a new centre needs to recalibrate the score before reading it as a probability or applying a threshold ([Supplementary section: calibration and thresholds]). Because similar AUCs at different centres can come from different evidence, we audited what each score used.

3.3. The evidence behind a score differs by cohort

When we audited what each score used†, the scores at MU depended mainly on where the patient was in treatment, and at UCSF on how much the tumour enhanced (Figure 3). At MU the scan schedule alone separates most progression scans from most non-progression scans, because most progression scans came more than 12 weeks after radiotherapy and most non-progression scans

193 came within those 12 weeks (Table 1). Across phases of care, adding the timeline to clinical data
194 and images gained 0.11 (0.04–0.20); within a phase it gained −0.02 (−0.13 to +0.11) (Figure 3A).
195 Within a phase of care, where the schedule cannot separate the scans, the image pathway gave
196 about as much evidence as the timeline, with AUCs of 0.68 (0.58–0.77) and 0.64 (0.51–0.78).

197 We built the image model on MU in three steps† to see which inputs raised its AUC (Figure 3C).
198 Adding change from earlier scans to the current scan's measurements gave the image pathway, 0.05
199 higher with an interval that included zero (0.00 to 0.11), and adding the three schedule inputs of
200 the planned imaging model (the interval since the previous scan, a flag for whether there was one
201 and the smallest volume counted from the end of radiotherapy) raised the AUC by a further 0.10
202 (0.05–0.15) across phases of care but not within them (−0.01, −0.06 to +0.04).

203 At UCSF the progression calls followed enhancement. Enhancing tumour volume alone, the
204 simplest baseline, gave an AUC of 0.88 (0.79–0.95), higher than the point estimates of the locked
205 image models; UCSF rated enhancement as increased at 27 of the 29 progression scans, and pro-
206 gression was dated a median of 7 days after the evaluated scan (Figure 3B). Among the 37 scans
207 rated as showing increased enhancement, the single-scan image model reached 0.65 (0.44–0.84), so
208 beyond that rating the measurements added little. At UCSF a model therefore scores well largely
209 by agreeing with a progression call that was itself made from enhancement at the same scan, which
210 is what the image pathway measures. Deep networks fine-tuned on UCSF scans learned above a
211 permuted-label control (Table 2), and their scores tracked enhancing volume† ([Supplementary sec-
212 tion: deep networks at MU and UCSF]). Because the evidence behind a score differs by cohort and
213 by situation, PRISM attaches a confidence to each score.

214 *3.4. The report states where its evidence is thin*

215 At the 80% level, the conformal sets of the combined score covered 78% to 87% of scans in the
216 three cohorts, meeting on point estimates the threshold fixed in advance, the nominal level less
217 two percentage points (Figure 4A). Most sets were informative, holding a single outcome for 62%
218 to 75% of scans. At UCSF the point estimate sat at the threshold and its interval reached below it
219 (68% to 87%). Coverage also differed by outcome†. The sets covered 77% (67% to 87%) of progression
220 scans at LUMIERE and 65% (51% to 79%) of non-progression scans at UCSF, against 94% and 97%
221 for the other outcome; on MU the two outcomes were similar. Thresholds fitted for each outcome
222 on MU did not correct this ([Supplementary section: system results]).

223 The warning that the two pathways disagree picked out the less accurate scans at MU. The
224 pathways disagreed on 40% of MU scans; the combined score's AUC was 0.90 where they agreed
225 and 0.59 where they disagreed, and setting the disagreeing scans aside gained more than removing
226 the same number matched for phase of care; both results met the tests fixed in advance (Table 2;
227 Figure 4B). Those tests used each cohort's own medians, and with them neither external cohort
228 met the tests; at LUMIERE the difference was 0.11 (−0.05 to 0.27). With the medians fixed on MU,
229 which the system uses because they can be computed at a single scan, the LUMIERE AUCs were
230 0.89 and 0.67 (difference 0.22, 0.10–0.34)†; at UCSF the interval still included zero. This rule flagged
231 60% of LUMIERE and 48% of UCSF scans ([Supplementary section: system results]).

232 The reliability card† states, for each scan, how many tested cases its situation holds, and this
233 differs by cohort. Across the three cohorts only four situations reached the tested tier, one at MU
234 and three at LUMIERE; 54 of the 146 MU scans fell in the tested situation, most LUMIERE scans

did, and every situation at UCSF had few cases (Figure 4C). At MU no situation in the first 12 weeks after radiotherapy held more than three progression scans. Tested situations had lower error than few-case situations at LUMIERE, with overlapping intervals (Table 2); at MU the order was reversed, because the few-case scans there were mostly non-progression scans early after radiotherapy, which the models score easily. The clinician reads these statements in the report, which we evaluated next.

3.5. The report passes an automatic check against its structured record

In the second report evaluation†, every report written by the local model passed the frozen lexical check of its numbers, flags and wording against its structured record (114, 162 and 130 reports at MU, LUMIERE and UCSF; 406 in total; Figure 5). The check cannot judge emphasis, order, readability or what a clinician would infer from the wording. Its records exclude the 10 LUMIERE and 10 UCSF patients used to develop the revision. The first evaluation had failed at LUMIERE (0 of 178) because one sentence quoted a number that was in no cited field; we revised the prompt and left the checker unchanged. In the second evaluation, all 220 reports requiring a few-case or untested tier stated it, and 28 of 30 perturbed records met the prespecified follows rule (Table 2; [Supplementary section: report v3.1]). Thus the report gives the clinician the segmentation figure and its volumes, separate pathway scores, confidence fields and the limits of the record, with each checked statement traceable to a record field. No clinician has yet evaluated whether this presentation helps a real surveillance decision.

4. Discussion

PRISM is a system for making longitudinal surveillance evidence inspectable. It gives the clinician separate image and treatment-timeline evidence at each scan, the measurements and context behind those fields, and a report written inside the hospital that states where the evidence is thin. The component models ranked scans at two external centres without refitting, but their evidence differed by setting. At the development centre the timeline provided most of the evidence, largely because of when patients were scanned; within a phase of care the images provided about as much. At UCSF enhancing volume provided the score, and enhancing volume alone did as well as the image pathway trained there. The locked results therefore establish transfer of ranking, not a portable risk scale: a group deploying PRISM at a new centre would recalibrate a probability and set any threshold on local scans.

Because its modules are separate, PRISM runs where data are incomplete. The image pathway needs only the scans and their masks, which is what most public surveillance datasets hold; at LUMIERE and UCSF it ran without treatment dates. Where dates exist, the timeline is added as a second pathway, and the combined score is shown beside both without changing the image model. A new cohort adds rows to the reliability card, so the tiers change as evidence accumulates and no confidence model has to be refitted.

Consider, as an illustration, a scan eight weeks after radiotherapy in which the enhancing tumour has grown. At MU, 1 of the 9 scans in this situation on the card (all grade 4 detection scans) was progression†. The timeline pathway scores such a scan low, because most scans at this point in treatment are not progression, and the image pathway scores it high, because enhancement has grown. PRISM's report would show both scores, state that they disagree, give a conformal set that

holds both outcomes and record that the card has few cases for this situation. The clinician then sees that the system has no measured basis for a call at this scan, in the weeks when pseudoprogression is most common, and decides with information PRISM does not use, such as symptoms, steroid dose and the radiotherapy plan.

Each of PRISM's components has been described before, and we are not aware of a system that combines them. Automatic segmentation of post-treatment glioma with measurement of change between scans is established (Rudie et al. 2022), conformal prediction has been applied to glioma segmentation (Elyassirad et al. 2025), clinical and imaging data have been fused to separate pseudoprogression from progression (Gomaa et al. 2025), and model cards report performance by subgroup (Mitchell et al. 2019). The closest system we found combines treatment variables extracted by a language model with tumour volumes to assign a structured follow-up category at one centre, without an uncertainty attached to the output (Jabal et al. 2026). On MU Christodoulou et al. (2026) predicted a label assigned at the patient level, on LUMIERE [cite Tikhonov 2025, arXiv:2509.06511; full reference not yet verified] classified each scan into four response categories, and Chen et al. (2026) predicted recurrence risk from preoperative and postoperative MRI across centres; none of these is detection at the scan.

PRISM was developed at one centre (Table 1), and the two external tests are its only evidence that its ranking of scans transfers; at UCSF its risk scale did not. The label is progression as each centre documented it; no scan was re-read for this study, and at UCSF most progression scans were also rated as showing increased enhancement. Subgroups are small, and the analyses marked † need testing on new data. No clinician has yet read the reports. PRISM forecast progression within 120 days less well than it detected it at the scan (AUC 0.60 in the intended use), so the forecast is a secondary task.

The next step is to pool the rare situations, early progression after radiotherapy and growing enhancement without it, across cohorts and train on them. The Burdenko and UCSD-PTGBM datasets add radiotherapy detail and advanced sequences, and their cases in the card's thin situations are where models fine-tuned on these long-tail cases could be tested. A reader study in which clinicians use PRISM's reports at the scan would show whether separate evidence and a stated confidence change their decisions.

Conclusion

PRISM keeps image and treatment-timeline evidence apart and shows the clinician both scores at every scan with how far each has been tested there; its models ranked scans at two other centres without refitting.

Acknowledgements

[PENDING AUTHOR DECISION].

Disclosure statement

[PENDING AUTHOR DECISION: competing interests, or "The authors report there are no competing interests to declare."] If any author serves on the editorial or advisory board of this journal,

state it here.

Declaration of generative AI use

[DRAFT, PENDING AUTHOR CONFIRMATION:] Generative AI is part of the system studied. PRISM's report generator uses a local language model (Qwen3.8-27B; six other open-weight models in a comparison); all ran offline on the authors' computing cluster and saw only derived system records. An AI assistant (Claude, Anthropic) was used to write and run analysis code and to draft and edit the manuscript under the authors' direction; it was given no patient-level data through any hosted service. The authors reviewed all code, results and text and take full responsibility for the content of this article.

Funding

No funding was obtained for this work.

Ethics statement

[PENDING AUTHOR DECISION: secondary analysis of de-identified datasets (MU-Glioma-Post and LUMIERE, public; UCSF-ALPTDG under a data use agreement); state the institutional determination or waiver and cite the source datasets' approvals.] No registered protocol. Analysis plans were committed to the repository before each external test; their commits are listed in Table 2. Patient and public involvement: none.

Data availability statement

[PENDING AUTHOR DECISION: MU-Glioma-Post (TCIA, doi:10.7937/7K9K-3C83) and LUMIERE (figshare; Suter et al. 2022) are public; UCSF-ALPTDG is available from its provider under a data use agreement; state which derived tables, fold assignments and locked model files are released, with a DOI.] Code: [PENDING: public repository link and release tag.]

Supplemental material

Supplemental data for this article are provided as a separate file.

Notes on contributors

[PENDING: a short biographical note for each author (no more than 200 words each).]

References

An C et al. 2021. Radiomics machine learning study with a small sample size: Single random training-test set split may lead to unreliable results. PLOS ONE. 16(8):e0256152. https://doi.org/10.1371/journal.pone.0256152

Angelopoulos AN, Bates S. 2023. Conformal prediction: A gentle introduction. Found Trends Mach Learn. 16(4):494–591. https://doi.org/10.1561/2200000101

Brandes AA, Franceschi E, Tosoni A et al. 2008. MGMT promoter methylation status can predict the incidence and outcome of pseudoprogression after concomitant radiochemotherapy in newly diagnosed glioblastoma patients. J Clin Oncol. 26(13):2192–2197. https://doi.org/10.1200/JCO.2007.14.8163

Chen C, Guo F, Huang D et al. 2026. Predicting recurrence risk of glioblastoma based on preoperative-postoperative longitudinal MRI: A multicenter study. Bioengineering. 13(6):668. https://doi.org/10.3390/bioengineering13060668

Christodoulou RC, Vamvouras G, Pitsillos R, Solomou EE, Georgiou MF. 2026. Explainable radiomics with probability calibration for postoperative glioblastoma surveillance. Eur J Radiol Artif Intell. 5:100074. https://doi.org/10.1016/j.ejrai.2026.100074

Elyassirad D, Gheiji B, Vatanparast M, Ahmadzadeh AM, Faghani S. 2025. CONSeg: Voxelwise uncertainty quantification for glioma segmentation using conformal prediction. Am J Neuroradiol. 46(12):2553–2560. https://doi.org/10.3174/ajnr.A8914

Fields BKK et al. 2024. The University of California San Francisco adult longitudinal post-treatment diffuse glioma MRI dataset. Radiol Artif Intell. 6(4):e230182. https://doi.org/10.1148/ryai.230182

Gomaa A, Huang Y, Stephan P et al. 2025. A self-supervised multimodal deep learning approach to differentiate post-radiotherapy progression from pseudoprogression in glioblastoma. Sci Reports. 15:17133. https://doi.org/10.1038/s41598-025-02026-7

Jabal MS et al. 2026. Agentic automation of BT-RADS scoring: End-to-end multi-agent system for standardized brain tumor follow-up assessment. [Preprint, arXiv:2603.21494]. `https://arxiv.org/abs/2603.21494`

Mahmoud E, Gass J, Dhemesh Y et al. 2025. MU-Glioma Post: A comprehensive dataset of automated MR multi-sequence segmentation and clinical features. Sci Data. 12(1):1847. https://doi.org/10.1038/s41597-025-06011-7

Mitchell M, Wu S, Zaldivar A et al. 2019. Model cards for model reporting. In: Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19). ACM. p 220–229

Rudie JD, Calabrese E, Saluja R et al. 2022. Longitudinal assessment of posttreatment diffuse glioma tissue volumes with three-dimensional convolutional neural networks. Radiol Artif Intell. 4(5):e210243. https://doi.org/10.1148/ryai.210243

Shi T, Kujawa A, Linares C, Vercauteren T, Booth TC. 2025. Automated longitudinal treatment response assessment of brain tumors: A systematic review. Neuro-Oncology. 27(8):1946–1971. https://doi.org/10.1093/neuonc/noaf037

Suter Y, Knecht U, Valenzuela W et al. 2022. The LUMIERE dataset: Longitudinal glioblastoma MRI with expert RANO evaluation. Sci Data. 9(1):768. https://doi.org/10.1038/s41597-022-01881-7

Taal W, Brandsma D, de Bruin HG et al. 2008. Incidence of early pseudo-progression in a cohort of malignant glioma patients treated with chemoirradiation with temozolomide. Cancer. 113(2):405–410. https://doi.org/10.1002/cncr.23562

Varoquaux G. 2018. Cross-validation failure: Small sample sizes lead to large error bars. NeuroImage. 180:68–77. https://doi.org/10.1016/j.neuroimage.2017.06.061

Tables

Table 1. The three cohorts. What each cohort records, and the scans in the detection task. Grade 4 glioma is the primary population.

	MU-Glioma-Post	LUMIERE	UCSF-ALPTDG
Role	Development (repeated cross-validation)	External, locked	External, locked
Access	Public	Public	Data use agreement
Scans per patient	Up to 6 index scans (median 2)	Up to 10 index scans (median 2)	One consecutive pair
Treatment dates	Surgery, radiotherapy start and end, dose, chemotherapy	None; rater flag for scans within three months of radiotherapy	Days from surgery; radiotherapy start only
Documented progression (the label)	Documented date in the clinical record	Expert response rating of each scan	Progression date supplied with the dataset
Segmentation masks	Automatic masks released with the dataset	DeepBraTumIA (three labels)	Masks provided with the dataset
Patients[a]	126	61	140
Age, median	62.5 (IQR 53–70)	62 (IQR 53–70)	57.5
Male	66 (52%)	32 (52%)	81 (58%)
IDH wildtype / mutant / unknown	112 / 8 / 6	38 / 1 / 22	81 / 13 / 46
MGMT methylated / unmethylated / unknown	45 / 62 / 19	26 / 28 / 7	71 / 43 / 26
Detection task[b]: progression / non-progression scans (patients)	62 / 84 (80) in the intended use; 64 / 117 (92) over all scans	73 / 105 (79)	29 / 40 (69), second scan only
Phase of care in the detection set	Progression: 56 of 62 later after radiotherapy; non-progression: 51 of 84 early after radiotherapy	Not recorded	65 of 69 later after radiotherapy[c]

[a] Patient characteristics describe the cohorts of the secondary 120-day task at MU-Glioma-Post (grade 4) and LUMIERE, and all grade 4 patients at UCSF-ALPTDG. Age is at diagnosis at MU-Glioma-Post and at surgery at LUMIERE.

[b] Positives are the scan at which progression was documented; negatives are defined per cohort in Section 2.2. The LUMIERE detection set has more patients than the 120-day cohort because it includes patients who contribute only a progression scan. At MU-Glioma-Post, the 146 scans of the intended use are the 181 grade 4 detection scans less the 35 taken before radiotherapy started, and 42 grade 4 progression scans of patients with no index scan (no scan with a 120-day label) are excluded. Reports were evaluated on the 114 intended-use scans of the 60 MU-Glioma-Post patients not used to develop the prompt, on 162 of the 178 LUMIERE scans and on 130 of the 140 grade 4 second scans at UCSF-ALPTDG, the rest belonging to development patients (Section 3.5).

[c] UCSF-ALPTDG gives only the first radiotherapy start; the phase assumes a six-week course. Two scans were early after radiotherapy and two had no radiotherapy date.

Table 2. Questions fixed in advance and their answers. Each rule was committed to the repository before the analysis it governs. AUC differences with 95% CIs.

Question	Rule fixed in advance	Result	Met
Detection on MU-Glioma-Post			
Do image measurements add to the timeline?	Planned model ≥ 0.75 (lower bound ≥ 0.65) and gain ≥ 0.03 with CI above 0	Model 0.837 (0.763–0.904); gain +0.015 (−0.032 to +0.063)	No
Do fine-tuned deep networks add?	Gain ≥ 0.03 with CI above 0; permuted-label control covers 0.5	No network added; control 0.483 (0.436–0.527)	No (control valid)
Does the locked model transport to LUMIERE?	External AUC ≥ 0.65 (below: transport failure)	AUC above 0.65 (Section 3.2); no transport failure	Yes
UCSF-ALPTDG			
Does the image pathway transport?	AUC ≥ 0.65, lower bound > 0.5	Both conditions met (Section 3.2)	Yes
Do images add to clinical data without a timeline?	Gain with CI above 0	+0.248 (+0.096 to +0.392)	Yes
Does training at UCSF-ALPTDG do better than the locked model?	Descriptive	Higher when trained there (Section 3.2)	Descriptive
Do deep networks learn at UCSF-ALPTDG?	Above the permuted-label control; control covers 0.5	+0.201 (+0.094 to +0.306) over a control of 0.541 (0.480–0.603)	Yes
The system (MU-Glioma-Post, LUMIERE, UCSF-ALPTDG)			
Do conformal sets keep their coverage?	Coverage ≥ 0.78 at the 80% level in each cohort	Combined score met on average in all three (by outcome, Section 3.4); images at LUMIERE 0.708 and timeline at UCSF-ALPTDG 0.696	Yes (combined)
Do flags pick out less accurate scans?	AUC unflagged minus flagged > 0, CI above 0	Disagreement at MU-Glioma-Post +0.307 (+0.175 to +0.432); per-cohort results in the supplement[a]	MU-Glioma-Post: yes (disagreement; set holding both outcomes). LUMIERE: set only. UCSF-ALPTDG: no
Does setting flagged scans aside beat matched removal?	Retained AUC minus matched removal > 0, CI above 0	Disagreement met at MU-Glioma-Post (Section 3.4); per-cohort results in the supplement[a]	MU-Glioma-Post: yes (disagreement). LUMIERE: no. UCSF-

Figure legends

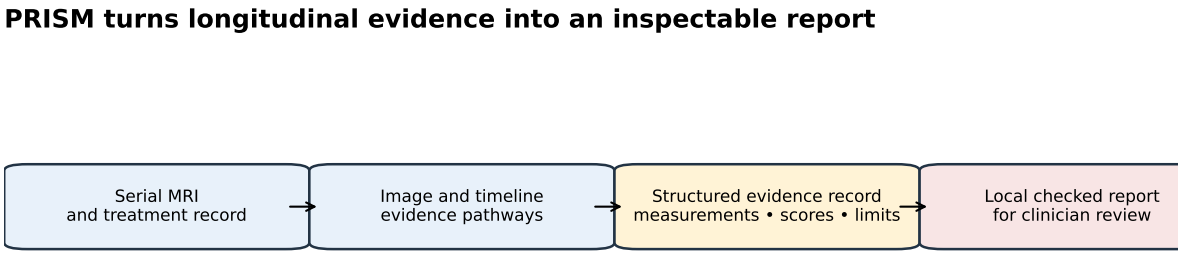


Figure 1. PRISM is built of modules, each answering a known challenge. A scan and its record enter two pathways: the timeline pathway (clinical data and the patient's position in treatment) and the image pathway (measurements from the segmentation masks supplied with each cohort). The combined score, one ridge model on both sets of inputs, is shown beside them. The confidence has three parts: a conformal prediction set at the 80% level, a statement of whether the two pathways agree, and the reliability card's tier for the scan's situation. A local language model writes the report from the record, and a deterministic checker tests its numbers, flags and confidence statements against the record. One example scan is followed through each module.

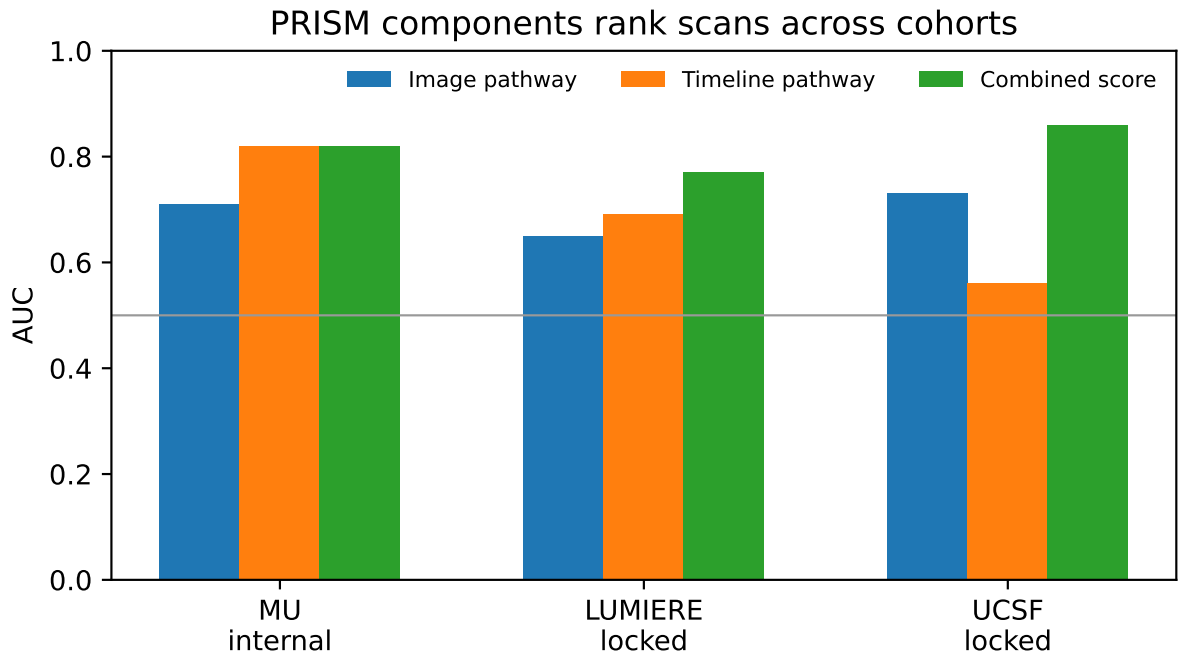


Figure 2. The models rank scans at two external centres without refitting. AUC for detecting the scan at which progression was documented, by pathway (clinical data alone, timeline, images, combined) and cohort: MU-Glioma-Post (repeated cross-validation, intended use); LUMIERE and UCSF-ALPTDG (models fitted once on MU-Glioma-Post and applied unchanged); and UCSF-ALPTDG with models trained there, as a reference. At LUMIERE and UCSF-ALPTDG the timeline is the reduced version, and the clinical-only model at LUMIERE is marked †. At UCSF-ALPTDG the images are shown for the single-scan model and for the image pathway with change features, which is the system's image pathway. The grey bar is the log enhancing tumour volume of the scan alone, with no fitted model†. Bars are 95% patient-bootstrap CIs.

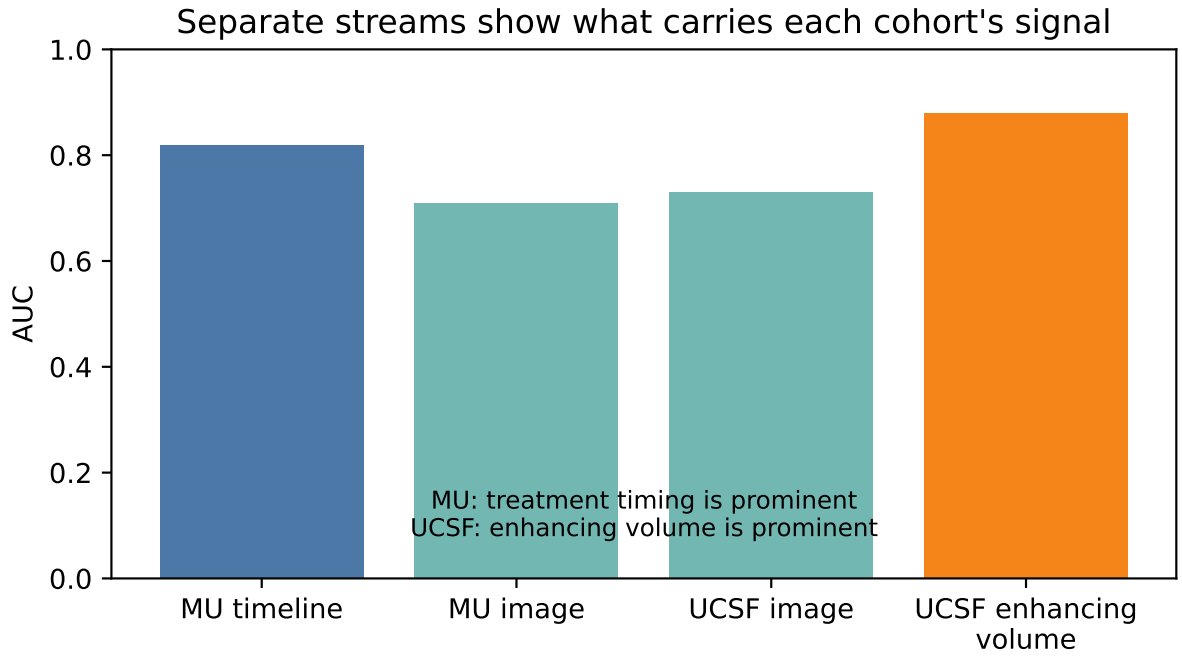


Figure 3. What each score rests on depends on the cohort. (A) MU-Glioma-Post: the gain from adding the timeline to clinical data and images, across phases of care and within them. (B) UCSF-ALPTDG: AUC of enhancing tumour volume alone beside the locked image models, and the share of progression and non-progression scans that UCSF rated as showing increased enhancement. (C) The image model on MU-Glioma-Post built in three steps (current scan; with change from earlier scans; with schedule inputs), across and within phases of care; the interval of the second step includes zero. All panels show analyses not fixed in advance (†). Bars are 95% patient-bootstrap CIs.

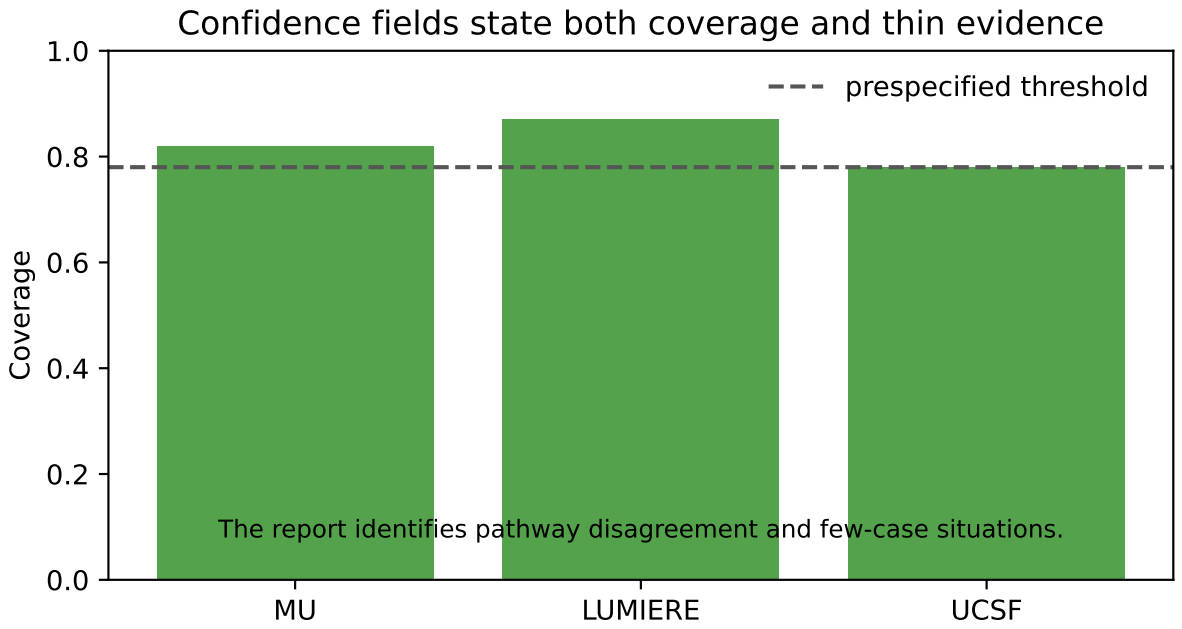


Figure 4. The confidence keeps its coverage on average, and disagreement marks the less accurate scans at MU-Glioma-Post. (A) Coverage of the 80% conformal sets by cohort and pathway; the dashed line is the threshold fixed in advance, met by the combined score on point estimates in all three cohorts; by outcome, coverage fell short for one outcome at LUMIERE and UCSF-ALPTDG (Section 3.4). (B) AUC of the combined score where the pathways agree and where they disagree, and after the disagreeing scans are set aside, compared with removal matched for phase of care, at MU-Glioma-Post. With each cohort's own medians, as fixed in advance, disagreement did not mark the less accurate scans at LUMIERE or UCSF-ALPTDG; with medians fixed on MU†, it did at LUMIERE. (C) The reliability card: situations (phase of care by change in enhancing tumour) in each cohort, shaded by tier (tested, few cases, not tested). The agreement rule and the card were defined after the cohort results had been seen (†).

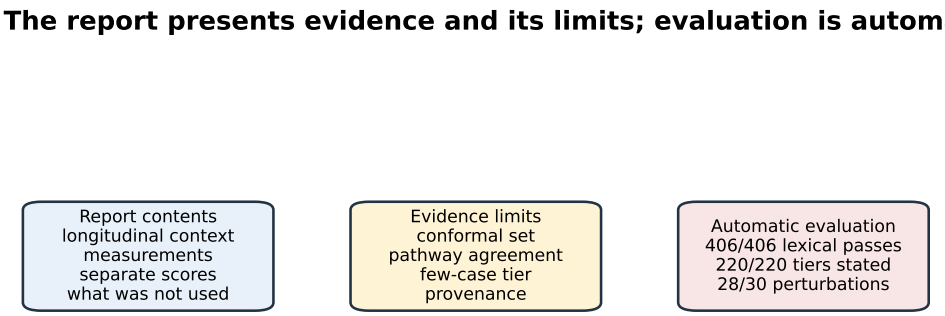


Figure 5. The report states the evidence and passes a check against the record. Left: one report as the clinician receives it, with the segmentation figure and its volumes, the two pathway scores, the combined score, the conformal set, the agreement statement and the card's tier. Right: share of reports passing the lexical check of numbers, flags and wording, by cohort, in the second evaluation, after one revision of the prompt following a failure at LUMIERE (Section 3.5).